

Project Proposal Preview

Request: add a search bar to my website

1. Requirements

Add a search bar to the website that allows users to search page content

Functional requirements

- Search input visible on all pages
- Submit via button or Enter key
- Display results on a dedicated page or modal
- Highlight matching terms in results
- Basic query parsing (case-insensitive, whitespace split)
- Show message when no results found

Non-functional requirements

- Responsive layout for mobile and desktop
- ARIA labels for accessibility
- Search response time <200ms
- No external JS libraries
- SEO-friendly markup
- Cross-browser compatibility (Chrome, Firefox, Safari, Edge)

Edge cases

- Empty query input
- Query with special characters
- Very long query string
- No matching results
- Network failure or JS disabled
- Duplicate results across pages

Assumptions

- Site content is static HTML
- Search will be implemented client-side
- No backend search API available
- Search index built at build time
- Users have JavaScript enabled
- Search limited to page content only

2. Architecture

Client-side search using a pre-built JSON index loaded at runtime

Component	Responsibility	Type
Search UI	Render input field, button, and ARIA attributes on every page	frontend
Search Index Builder	Traverse static HTML at build time and generate a compact search-index.json	integration

Component	Responsibility	Type
Search Engine	Load index.json, parse queries (case-insensitive, whitespace split), and find matching pages	frontend
Results Renderer	Display matches in a modal or dedicated results page, highlight terms, handle no-result and error states	frontend

Data flow

- Build step: run Search Index Builder to produce search-index.json
- Page load: Search Engine fetches search-index.json asynchronously
- User types query and presses Enter or clicks Search button
- Search Engine parses query and searches the in-memory index
- Results Renderer receives match list, highlights query terms, and shows them in modal or redirects to results page
- If no matches, Renderer shows a friendly 'no results' message
- If fetch fails, Renderer shows an error fallback

File structure

- build/search-index.js
- public/search-index.json
- assets/js/search.js
- assets/css/search.css
- templates/search-modal.html
- templates/search-results.html

3. Tools

Tool	Category	Purpose	Required
Node.js	backend	Runtime for build scripts and index generation	Yes
npm	backend	Package manager for dependencies like cheerio	Yes
cheerio	integration	Parse static HTML during build to create search-index.json	Yes

4. Cost Estimate (tools/services only)

Tool	Pricing model	Est. cost	Notes
Node.js	free	\$0	Open source runtime, free to use
npm	free	\$0	Package manager, free
cheerio	free	\$0	NPM package, free

Estimated monthly total: \$0

Note: These are third-party tool/service cost estimates only, not a cost of building the feature